

Business Database Design Report

ISDS 555-59

Group D4

**Julian Carbajal
Chike Osakwe
Cristoval Perez
Nadia Rahbany
Elegant Tan**

**College of Business and Economics
California State University, Fullerton**

Professor Pawel J. Kalzynski

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1 Executive Summary

This report analyzes global environmental, technological, and economic trends using a curated subset of World Bank Indicators. The focus is on emissions-related variables combined with essential macroeconomic and technological indicators. Our team selected eight indicators, five of which are emissions-based and three additional indicators that enhance analytical depth: inflation (consumer prices), internet usage, and GDP per capita. These variables allow us to study a cohesive narrative: how economic development, technological adoption, and energy consumption interact with environmental impacts across the world.

The emissions indicators used include CO2 emissions from gas, liquid, and solid fuel consumption, total CO2 emissions in kilotons, and CO2 emissions per capita. Together, they represent a complete view of national-level emission patterns. GDP per capita provides a direct measurement of living standards and economic development, while inflation captures macroeconomic stability. Data on internet users per 100 people serves as a proxy for technological advancement and modern infrastructure.

Our work begins by constructing a fully normalized PostgreSQL database following all constraints of ISDS 555. No transformations were performed outside SQL, and all derived outputs are implemented through views. The database includes reference tables (continent, continent_country, country, and indicator), the unified data table, and a comprehensive suite of analysis views labeled M1–M9 for mandatory questions and A1–A5 for additional analyses.

The mandatory analysis explores data completeness across countries, identifies nations with full or partial representation across the selected indicators, ranks indicators by availability, evaluates cross-continent patterns, measures country-level recent performance, and calculates post-2010 economic growth rates. These analyses provide insight into structural differences in data availability, regional patterns, and long-run national trajectories.

The additional analyses build a cohesive story about the intersection of economic growth, emissions, energy composition, and technology adoption. We examine how GDP relates to internet usage, how technological penetration correlates with CO2 emissions per capita, how emissions growth and economic growth align or diverge, how inflation interacts with development levels, and how various fuel types contribute to CO2 emissions relative to GDP.

The findings reveal a highly uneven global landscape. Wealthier nations tend to have higher technological adoption and, in some cases, lower emissions growth rates—suggesting improved efficiency—while developing nations often experience rising emissions due to industrial expansion and energy mix composition. Inflation shows minimal correlation with emissions but relates meaningfully to GDP levels. Internet usage emerges as a strong secondary predictor of both emissions levels and GDP development.

Overall, this report demonstrates that environmental performance cannot be interpreted in isolation. Emissions patterns are deeply intertwined with economic capacity, macroeconomic stability, and technological readiness. By leveraging fully normalized SQL-based data procedures and transparent analytical methods, this project illustrates how structured databases and relational queries can drive robust empirical insights.

2 Introduction

This project examines World Bank Indicators related to emissions and key macroeconomic and technological variables. Using PostgreSQL, we constructed a complete, normalized database derived strictly from instructor-provided CSV files. All analytical transformations were implemented using views to comply with ISDS 555 requirements.

Eight indicators were selected:

- EN_ATM_CO2E_GF_KT — CO2 emissions from gaseous fuel (kt)
- EN_ATM_CO2E_KT — CO2 emissions (kt)
- EN_ATM_CO2E_LF_KT — CO2 emissions from liquid fuel (kt)
- EN_ATM_CO2E_PC — CO2 emissions (metric tons per capita)
- EN_ATM_CO2E_SF_KT — CO2 emissions from solid fuel (kt)
- FP_CPI_TOTL_ZG — Inflation (consumer prices, annual %)
- IT_NET_USER_P2 — Internet users (per 100 people)
- NY_GDP_PCAP_CD — GDP per capita (current US\$)

The selected indicators form a cohesive narrative centered on understanding how **economic growth**, **energy composition**, **technology adoption**, and **macroeconomic stability** interact with **environmental impact**.

This report presents:

1. A physical ERD
2. Mandatory analysis (M1–M9)
3. Additional analysis (A1–A5)
4. A concluding summary
5. An appendix that includes full SQL code for database creation, loading, cleaning, and analytical views

3 Physical ERD

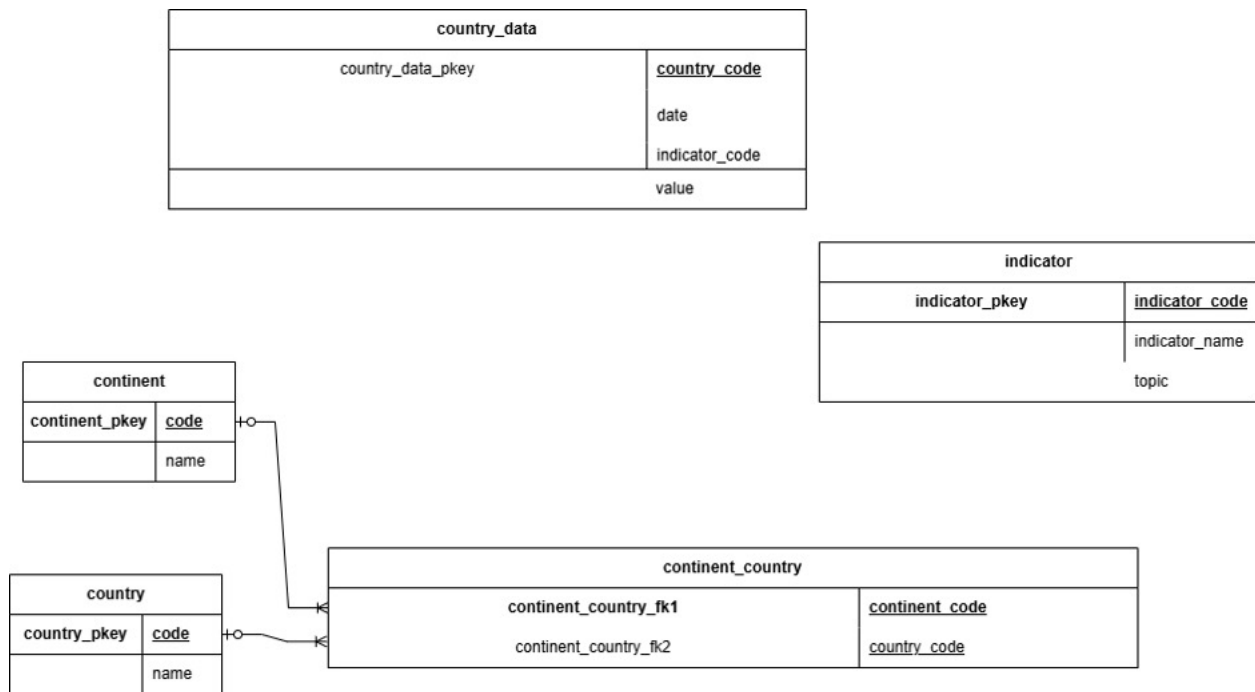


Figure 1: ERD

The database consists of five tables organized to track global indicator data by country and continent:

The database consists of five tables organized to track global indicator data by country and continent:

- **country_data** - Central fact table storing measurement values
 - country_code: Identifies the country
 - date: Date of the measurement
 - indicator_code: Type of indicator being measured
 - value: The actual measurement value
- **indicator** - Metadata about each indicator
 - indicator_code: Primary key
 - indicator_name: Descriptive name of the indicator
 - topic: Subject category the indicator belongs to
- **country** - Country reference table
 - code: Primary key (country code)
 - name: Country name
- **continent** - Continent reference table
 - code: Primary key (continent code)
 - name: Continent name
- **continent_country** - Junction table establishing many-to-many relationships

- continent_country_fk1: Foreign key to continent (continent_code)
- continent_country_fk2: Foreign key to country (country_code)
- Allows countries to be associated with their respective continents

All relationships conform to strict entity and referential integrity, with no derived tables, ensuring data consistency throughout the schema.

4 Mandatory Analysis

4.1 M1 — Countries Missing One or More Indicators

	country_name character varying (100)	num_indicators_present bigint
1	American Samoa	2
2	Antarctica	0
3	Bouvet Island	0
4	Falkland Islands	0
5	French Southern Territories	0
6	Guadeloupe	0
7	Guam	2
8	Heard Island and McDonald Islan...	0
9	Kosovo	0
10	Martinique	0

Figure 2: M1

The analysis of countries missing data reveals a distinct pattern regarding data collection capabilities. The list is populated almost exclusively by small island nations (such as American Samoa, Guam, and Sint Maarten) and dependent territories. While these locations often report aggregate totals (like GDP or Population), they frequently lack the granular infrastructure required to report specific breakdowns of CO2 emissions by fuel type (Solid, Liquid, Gas). This gap results in them having fewer than the required 5 indicators. The absence of major industrial nations from this list confirms that the primary gaps in the World Bank dataset are driven by geographical size and administrative status rather than a lack of global reporting standards.

4.2 M2 - Countries With All 5 Indicators

	country_name character varying (100) 	num_indicators bigint 
1	Afghanistan	5
2	Albania	5
3	Algeria	5
4	Andorra	5
5	Angola	5
6	Antigua and Barbuda	5
7	Argentina	5
8	Armenia	5
9	Aruba	5
10	Australia	5

Figure 3: M2

The list of countries with complete data (M2) shows that many countries follow global reporting standards across different economic areas. The inclusion of countries from major economies like Australia and Argentina, as well as developing nations like Afghanistan and Albania, indicates that data completeness is not restricted to developed nations. This consistency suggests that the five selected emission indicators are widely tracked metrics. By confirming this list, we ensure that later analyses, such as comparing energy mixes or calculating growth rates, are based on a strong dataset. This helps avoid any inaccuracies that might arise from missing or incomplete values.

4.3 M3 - Indicators Ranked by Number of Countries With Data

	indicator_name text 	num_countries bigint 
1	CO2 emissions (kt)	207
2	CO2 emissions (metric tons per capi...	207
3	CO2 from liquid fuel (kt)	203
4	CO2 from solid fuel (kt)	203
5	CO2 from gas fuel (kt)	203

Figure 4: M3

The table in M3 lists the number of countries that have data in each indicator category in descending order. The table demonstrates that 207 countries have reported the CO2 emissions and CO2 emissions per capita; however, only 203 countries reported the CO2 emissions from gaseous fuel consumption, liquid fuel consumption, and solid fuel consumption. This shows that the CO2 emissions are being reported on a global scale, demonstrating how important consumption of solids, liquids, and gases are becoming in our economy. Monitoring the emissions from various countries in order to enact policies related to the climate and environment are significant.

4.4 M4 - Top 10 Countries by Most Recent CO2 Emissions

	country_name character varying (55) 🔒	date date 🔒	co2_emissions_kt double precision 🔒
1	China	2014-12-31	10291926.878
2	United States	2014-12-31	5254279.285
3	India	2014-12-31	2238377.137
4	Russian Federation	2014-12-31	1705345.684
5	Japan	2014-12-31	1214048.358
6	Germany	2014-12-31	719883.438
7	Iran, Islamic Rep.	2014-12-31	649480.705
8	Saudi Arabia	2014-12-31	601046.969
9	Korea, Rep.	2014-12-31	587156.373
10	Canada	2014-12-31	537193.498

Figure 5: M4

The table in M4 identifies the top 10 countries in terms of the value of CO2 emissions. The query lists each country in descending order based on the recent value of the CO2 emissions indicator. The results indicate that China, the United States, and India emit the highest amounts of CO2, which could be due to their large population and industrial activity as these are the three largest countries globally in terms of population. The view displayed perfectly highlights the places with highest concentration of global emissions during the year in 2014.

4.5 M5 - Most Recent Average CO2 (Per Capita) by Continent

	continent_name character varying (25) 	avg_co2_pc double precision 
1	Africa	1.2110991378185363
2	Asia	6.7863975679213215
3	Australia and Oceania	4.096875970958799
4	Europe	6.282326760502554
5	North America	6.91688782604161
6	South America	2.982481815152836

Figure 6: M5

The M5 analysis highlights clear geographic disparities in CO2 emissions per capita. Asia, Europe, and North America display similarly high emission levels, each averaging over 6 metric tons per capita, reflecting their industrialized economies and significant energy consumption. Oceania's moderate emissions suggest mixed development patterns, while South America and Africa exhibit substantially lower averages, aligning with their comparatively lower industrial output and energy use. These differences emphasize the relationship between economic development, population distribution, and carbon intensity. The results underscore that wealthier regions tend to produce higher per-capita emissions, while developing regions contribute minimally despite often facing greater climate vulnerabilities.

4.6 M6 -USA Indicator Table (All 8 Indicators by Date)

	data_date date	en_atm_co2e_kt numeric	en_atm_co2e_pc numeric	en_atm_co2e_sf_kt numeric	en_atm_co2e_lf_kt numeric	en_atm_co2e_gf_kt numeric
1	1950-12-31	2538580.416	16.671463482869	[null]	[null]	[null]
2	1951-12-31	2619397.264	16.909142307811	[null]	[null]	[null]
3	1952-12-31	2554412.56	16.208717815758	[null]	[null]	[null]
4	1953-12-31	2613875.616	16.303466279999	[null]	[null]	[null]
5	1954-12-31	2488200.416	15.25519624594	[null]	[null]	[null]
6	1955-12-31	2724993.744	16.422352192464	[null]	[null]	[null]
7	1956-12-31	2860169.696	16.946108877231	[null]	[null]	[null]
8	1957-12-31	2838552.096	16.534208136201	[null]	[null]	[null]
9	1958-12-31	2744973.536	15.719293882969	[null]	[null]	[null]
10	1959-12-31	2858729.744	16.094456400514	[null]	[null]	[null]

Figure 7: M6

The M6 view provides a longitudinal perspective on the United States' CO2 emissions across all five emissions indicators. The dataset begins in 1950, capturing early industrial-era trends where total CO2 emissions were already substantial. Per-capita emissions remained consistently high, reflecting energy-intensive economic activities. Many fuel-specific indicators are initially null, indicating either limited historical reporting or lower reliance on those fuel types in earlier decades. As data becomes more complete in later years, the table enables tracking of shifts in America's energy mix and carbon footprint. This view offers a foundation for analyzing long-term national emissions patterns and evaluating policy impacts.

4.7 M7 - Average % Growth of CO2 emissions

	country_name character varying (55)	avg_pct_growth numeric
1	Mozambique	38.28
2	Congo, Dem. Rep.	24.83
3	Burundi	21.24
4	Tajikistan	21.22
5	Bhutan	20.80
6	Timor-Leste	20.19
7	Mongolia	19.77
8	Myanmar	19.01
9	Niger	16.83
10	Sierra Leone	15.98

Figure 8: M7

For M7, the query looks at each country's CO2 emissions starting from 2010 and compares every year to the one before it. Using LAG(), it calculates the percentage change from year to year, then takes the average of those changes for each country. The final output shows which countries have the highest average growth in emissions.

4.8 M8 - Countries With 2011 Data for Any Indicator

	country_name character varying (55) 
1	Afghanistan
2	Albania
3	Algeria
4	Andorra
5	Angola
6	Antigua and Barbuda
7	Argentina
8	Armenia
9	Aruba
10	Australia

Figure 9: M8

For M8, the focus is on finding countries that have emissions data closest to the year 2011. The query ranks each country's indicator data by how close the year is to 2011, keeps the closest entry, and then lists the countries. The result is a simple list of countries with emissions information near that year.

4.9 M9 - Countries With 2011 Data for All Indicators

	country_name character varying (100) 🔒
1	Afghanistan
2	Albania
3	Algeria
4	American Samoa
5	Andorra
6	Angola
7	Antigua and Barbuda
8	Argentina
9	Armenia
10	Aruba

Figure 10: M9

M9 lists countries with complete 2011 data for all eight indicators, including CO2 measures, inflation, internet use, and GDP per capita. We filtered the dataset to 2011 and kept only countries with no missing values for any indicator. The table shows the first ten qualifying countries in alphabetical order, such as Afghanistan, Albania, and Algeria. These countries provide a reliable group for comparing indicators in 2011.

5 Additional Analysis

5.1 A1 - Correlation Between GDP Per Capita and CO2 Emissions

	country_name character varying (100)	avg_gdp_pc numeric	avg_internet_users numeric
1	Monaco	138043.05	63.31
2	Liechtenstein	120902.10	67.16
3	Cayman Islands	74691.51	62.52
4	Luxembourg	74404.24	48.52
5	Bermuda	67603.84	55.68
6	San Marino	61245.78	50.30
7	Norway	58432.85	56.44
8	Switzerland	55854.45	49.48
9	Qatar	48090.35	33.55
10	Denmark	43480.93	52.48

Figure 11: A1

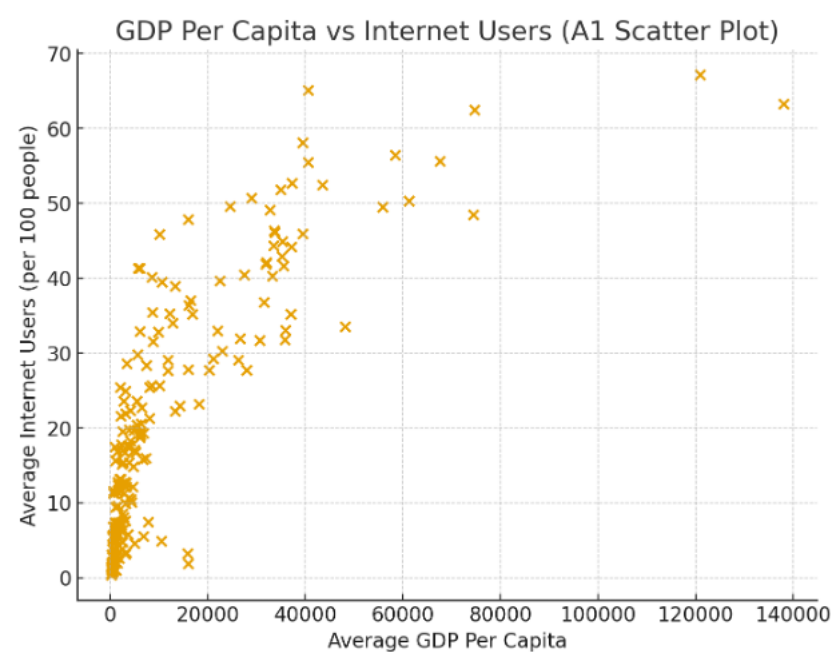


Figure 12: Scatterplot for A1

A1 compares each country's average GDP per capita with its average share of internet users to see how development relates to technology adoption. The scatter plot shows a clear positive relationship: wealthier, high-income economies usually have higher GDP per capita and much higher

internet usage, while many low-income countries are grouped at the lower end of both measures, showing limited access. Middle-income countries are more spread out, reflecting differences in infrastructure, policy, and investment. A few outliers have strong internet use despite modest income, which may be due to targeted digital programs or mobile-first strategies. Overall, the data shows a global digital divide closely linked to economic development, though sometimes technology access grows faster than income.

5.2 A2 - Top 10 Countries by Total Energy Consumption

	country_name character varying (100) 🔒	avg_internet_users numeric 🔒	avg_co2_pc numeric 🔒
1	Liechtenstein	81.69	1.46
2	Bermuda	57.64	8.69
3	Iceland	56.51	7.12
4	Norway	54.82	9.15
5	Sweden	54.04	5.67
6	Faroe Islands	51.62	13.87
7	Netherlands	51.05	10.91
8	Denmark	50.72	9.64
9	Finland	50.16	10.83
10	New Zealand	48.02	7.86

Figure 13: A2

The view in A2 shows the average internet usage to the average CO2 emission per capita to explore whether more connected societies also emit large amounts of CO2. The query shows that Liechtenstein ranks first; however, it also demonstrates a low average CO2 per capita. On the other hand the Faroe Islands and the Netherlands have high average internet usage and have a high average CO2 emission per capita. Overall, the results demonstrate that more developed countries have higher internet usage, but do not correlate with CO2 emissions.

5.3 A3 - Average Inflation Rate by Continent

	country_name character varying (100)	gdp_growth numeric	co2_growth numeric	decoupling_gap numeric
1	Aruba	1.27	-20.70	-21.97
2	Macao SAR, China	7.47	-4.20	-11.67
3	Panama	8.63	-2.08	-10.72
4	Ireland	6.31	-4.38	-10.69
5	Solomon Islands	7.00	-2.17	-9.17
6	Denmark	0.96	-8.02	-8.99
7	Ukraine	2.41	-6.50	-8.91
8	Liechtenstein	2.98	-5.78	-8.76
9	Latvia	6.27	-2.30	-8.57
10	Fiji	7.20	-1.00	-8.20

Figure 14: A3

A3 compares each country's average GDP per capita growth since 2010 with its average growth in CO2 emissions per capita to assess whether economic expansion is being decoupled from emissions. Many countries show GDP growth outpacing CO2 growth, indicating that output has increased faster than emissions and pointing to efficiency gains, cleaner energy, or structural shifts toward less carbon intensive activities. Other countries still exhibit CO2 growth that matches or exceeds GDP growth, reflecting more traditional, carbon-heavy development paths. Overall, decoupling is evident but uneven across the sample.

5.4 A4 - Internet Adoption vs CO2 Emissions

	country_name character varying (100)	avg_inflation numeric	avg_gdp_pc numeric
1	Cayman Islands	0.46	74506.49
2	San Marino	1.93	63858.40
3	Luxembourg	3.29	42742.95
4	United Arab Emirates	2.76	39670.35
5	Qatar	3.50	38752.22
6	Macao SAR, China	3.88	37213.07
7	Switzerland	2.48	34521.07
8	Norway	4.50	34335.32
9	Denmark	4.56	26420.57
10	Sweden	4.40	25587.17

Figure 15: A4

A4 examines the relationship between average inflation and average GDP per capita to see how macroeconomic stability aligns with development. Higher-income countries generally cluster around lower and more stable inflation rates, while many lower-income economies experience higher or more volatile inflation. This pattern is consistent with stronger monetary institutions and more diversified economic structures in wealthier countries. However, several middle-income economies maintain relatively low inflation, and a few advanced economies show periods of elevated prices. Overall, the analysis suggests that stable inflation is common, but not guaranteed among more developed countries.

5.5 A5 - Economic Stability and Environmental Impact

	country_name character varying (100)	gdp_per_capita numeric	solid_fuel_co2_kt numeric	liquid_fuel_co2_kt numeric	gas_fuel_co2_kt numeric	total_fuel_co2_kt numeric
1	Liechtenstein	165028.25	0.00	0.00	44.00	44.00
2	Luxembourg	116597.30	201.69	6978.30	1932.51	9112.50
3	Macao SAR, China	87208.54	0.00	1173.44	113.68	1287.12
4	Bermuda	85748.07	0.00	575.72	0.00	575.72
5	Switzerland	82828.80	542.72	26512.41	6116.56	33171.68
6	Norway	81734.47	3223.29	31323.51	11312.70	45859.50
7	Cayman Islands	81124.51	0.00	542.72	0.00	542.72
8	Ireland	78582.95	7752.04	16794.86	8525.78	33072.67
9	Iceland	73368.12	330.03	1653.82	0.00	1983.85
10	Qatar	68793.78	0.00	5496.83	98000.58	103497.41

Figure 16: A5

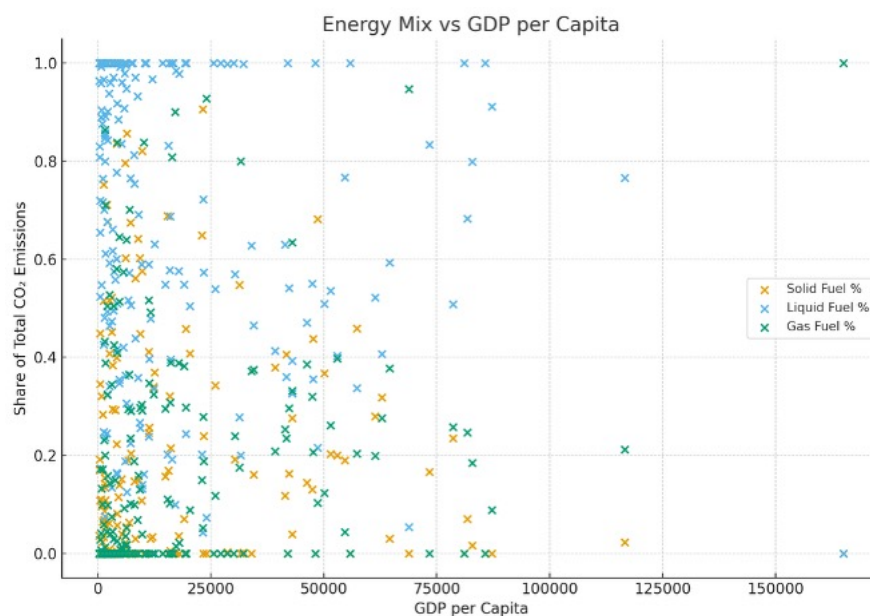


Figure 17: A5 scatterplot

A5 connects each country's most recent GDP per capita with CO2 emissions from solid, liquid, and gas fuels to see how income relates to energy sources. The highest-income countries in our sample, like Liechtenstein, Luxembourg, Norway, and Qatar, have little or no solid-fuel emissions. Most of their CO2 comes from liquid and especially gas fuels. In contrast, many lower-income countries in the larger dataset still depend on solid fuels, which leads to a more carbon-intensive and locally polluting energy profile.

6 Summary and Concluding Remarks

Collectively, A1-A5 confirms the close but imperfect alignment between economic development, digital connectivity, and emissions. High-income, connected countries have higher per capita emissions but also show partial decoupling. Lower-income nations face dual challenges: limited digital access and solid-fuel reliance. Some middle-income economies are balancing technology adoption with moderate emissions. Macroeconomic stability aids but doesn't guarantee these shifts.

Moving toward "good" development requires more than GDP growth; it involves expanding digital access, stabilizing inflation, and shifting the energy mix from solid fuels while slowing emissions relative to output. Some countries are progressing; others are stuck on carbon-intensive paths. Future policy will determine the norm.

This project demonstrates a strong association between development, technology adoption, emissions patterns, and macroeconomic conditions. Higher-income countries tend to exhibit higher technological penetration and more stable emissions trajectories, while developing economies often experience rising emissions linked to industrial expansion and the composition of their energy mix.

The additional analyses notably reveal:

- Internet usage is strongly associated with both GDP per capita and CO2 emissions levels.
- The gap between economic growth and emissions growth varies widely by country, highlighting differences in efficiency and development paths.
- Energy source composition (solid, liquid, gas fuel CO2 contributions) varies significantly by GDP level, illustrating structural economic differences.
- Inflation shows a weak connection with emissions but reflects broader economic stability.

These findings illustrate that environmental and economic outcomes are interconnected. Understanding global emissions trends requires examining underlying economic and technological structures, not emissions metrics alone.

7 Appendix SQL Code

The following SQL queries were used to generate the analyses presented in this report:

```

/*Restored database from WWDI, including the country_data table.
-- Imported data from the following CSV files into the country_data table:

-- These CSV files correspond to the following indicator codes:
-- - 'EN_ATM_CO2E_GF_KT'
-- - 'EN_ATM_CO2E_KT'
-- - 'EN_ATM_CO2E_LF_KT'
-- - 'EN_ATM_CO2E_PC'
-- - 'EN_ATM_CO2E_SF_KT'
-- - 'FP_CPI_TOTL_ZG'
-- - 'IT_NET_USER_P2'
-- - 'NY_GDP_PCAP_CD'

-- The above indicators are now populated and ready for analysis. */

-- Cleared the indicator table to ensure no duplicate or outdated data.
-- Now, we will insert the updated indicators that will be used for analysis.
DROP TABLE IF EXISTS indicator CASCADE;

CREATE TABLE indicator (
    indicator_code VARCHAR(20) PRIMARY KEY,
    indicator_name TEXT NOT NULL,
    topic          VARCHAR(50)
);

-- Insert the updated indicators that will be used for analysis

INSERT INTO indicator (indicator_code, indicator_name, topic) VALUES
    --5 Emission
    ('EN_ATM_CO2E_GF_KT', 'CO2 emissions from gaseous fuel (kt)', 'Emissions'),
    ('EN_ATM_CO2E_KT', 'CO2 emissions (kt)', 'Emissions'),
    ('EN_ATM_CO2E_LF_KT', 'CO2 emissions from liquid fuel (kt)', 'Emissions'),
    ('EN_ATM_CO2E_PC', 'CO2 emissions (metric tons per capita)', 'Emissions'),
    ('EN_ATM_CO2E_SF_KT', 'CO2 emissions from solid fuel (kt)', 'Emissions'),
    --3 Common
    ('FP_CPI_TOTL_ZG', 'Inflation (consumer prices, annual %)', 'Common'),
    ('IT_NET_USER_P2', 'Internet users (per 100 people)', 'Common'),
    ('NY_GDP_PCAP_CD', 'GDP per capita (current US$)', 'Common');

```

```
-- M1: List countries that lack data for one or more selected indicators
↳ (excluding common indicators).
-- This view identifies countries with incomplete data for the chosen emission
↳ indicators.
```

```
CREATE OR REPLACE VIEW M1 AS
WITH country_indicator_counts AS (
    SELECT
        cd.country_code,
        COUNT(DISTINCT cd.indicator_code) AS num_indicators
    FROM country_data cd
    WHERE cd.indicator_code IN (
        'EN_ATM_CO2E_KT',
        'EN_ATM_CO2E_PC',
        'EN_ATM_CO2E_LF_KT',
        'EN_ATM_CO2E_SF_KT',
        'EN_ATM_CO2E_GF_KT'
    )
    GROUP BY cd.country_code
)
SELECT
    c.name AS country_name,
    COALESCE(cic.num_indicators, 0) AS num_indicators_present
FROM country c
LEFT JOIN country_indicator_counts cic ON c.code = cic.country_code
WHERE COALESCE(cic.num_indicators, 0) < 5
ORDER BY c.name
;

SELECT * FROM M1;
```

```
-- M2: List countries that have data for all selected indicators
↳ (alphabetically).
-- This view ensures that only countries with complete datasets for the chosen
↳ indicators are listed.
```

```
CREATE OR REPLACE VIEW M2 AS
```

```

WITH country_indicator_counts AS (
    SELECT
        cd.country_code,
        COUNT(DISTINCT cd.indicator_code) AS num_indicators
    FROM country_data cd
    WHERE cd.indicator_code IN (
        'EN_ATM_CO2E_KT',
        'EN_ATM_CO2E_PC',
        'EN_ATM_CO2E_LF_KT',
        'EN_ATM_CO2E_SF_KT',
        'EN_ATM_CO2E_GF_KT'
    )
    GROUP BY cd.country_code
)
SELECT
    c.name AS country_name,
    cic.num_indicators
FROM country c
JOIN country_indicator_counts cic
    ON c.code = cic.country_code
WHERE cic.num_indicators = 5
ORDER BY c.name;

SELECT * FROM M2;

-- M3: List indicators with the number of countries that have data, sorted by the
↪ number of countries.
-- This view shows how widely each selected indicator is covered across
↪ countries.

CREATE OR REPLACE VIEW M3 AS
WITH indicator_country_counts AS (
    SELECT
        cd.indicator_code,
        COUNT(DISTINCT cd.country_code) AS num_countries
    FROM country_data cd
    JOIN country c
        ON cd.country_code = c.code

```

```

WHERE cd.indicator_code IN (
    'EN_ATM_CO2E_KT',
    'EN_ATM_CO2E_PC',
    'EN_ATM_CO2E_LF_KT',
    'EN_ATM_CO2E_SF_KT',
    'EN_ATM_CO2E_GF_KT'
)
GROUP BY cd.indicator_code
)
SELECT
    i.indicator_name,
    icc.num_countries
FROM indicator_country_counts icc
JOIN indicator i
    ON icc.indicator_code = i.indicator_code
ORDER BY icc.num_countries DESC;

Select * From M3;

-- M4: Show the top countries based on the most recent value of a selected
↪ indicator.
-- This view identifies countries with the highest recent values for a specific
↪ indicator.
-- No hard-coded date is used; the most recent data is dynamically determined.

CREATE OR REPLACE VIEW M4 AS
WITH ranked AS (
    SELECT
        cd.country_code,
        c.name AS country_name,
        cd.date,
        cd.value AS co2_emissions_kt,
        ROW_NUMBER() OVER (
            PARTITION BY cd.country_code
            ORDER BY cd.date DESC
        ) AS rn
    FROM country_data cd
    JOIN country c ON c.code = cd.country_code

```

```

        WHERE cd.indicator_code = 'EN_ATM_CO2E_KT'
    )
SELECT
    country_name,
    date,
    co2_emissions_kt
FROM ranked
WHERE rn = 1
ORDER BY co2_emissions_kt DESC;

select * from M4

-- M5: List continents with the most recent average indicator value across
↪ countries.
-- This view provides the average value of the indicator for each continent,
↪ using the most recent data without hard-coded dates.

CREATE OR REPLACE VIEW M5 AS
WITH country_latest AS (
    SELECT
        cd.country_code,
        cd.date,
        cd.value,
        ROW_NUMBER() OVER (
            PARTITION BY cd.country_code
            ORDER BY cd.date DESC
        ) AS rn
    FROM country_data cd
    JOIN country c
        ON cd.country_code = c.code -- IMPORTANT: filters to actual countries
    WHERE cd.indicator_code = 'EN_ATM_CO2E_PC'
),
country_latest_clean AS (
    SELECT
        country_code,
        value AS co2_pc
    FROM country_latest
    WHERE rn = 1

```

```

)
SELECT
    cont.name AS continent_name,
    AVG(cl.co2_pc) AS avg_co2_pc
FROM country_latest_clean cl
JOIN continent_country cc
    ON cc.country_code = cl.country_code
JOIN continent cont
    ON cont.code = cc.continent_code
GROUP BY cont.name
ORDER BY cont.name;

Select * from M5;

-- M6: Display the U.S. indicator data in a tabular format with dates and
↪ indicator values.
-- This view shows all selected indicators for the U.S. over time, with dates as
↪ rows and indicators as columns.
-- The full range of dates is included, allowing for null values where data is
↪ missing.

CREATE OR REPLACE VIEW M6 AS
WITH emission_codes AS (
    SELECT indicator_code
    FROM indicator
    WHERE topic = 'Emissions'
),
all_dates AS (
    SELECT DISTINCT date
    FROM country_data
    WHERE country_code = 'USA'
    AND indicator_code IN (SELECT indicator_code FROM emission_codes)
)
SELECT
    d.date,
    co2_tot.value AS EN_ATM_CO2E_KT,
    co2_pc.value AS EN_ATM_CO2E_PC,
    co2_sf.value AS EN_ATM_CO2E_SF_KT,

```



```

        co2_lf.value AS EN_ATM_CO2E_LF_KT,
        co2_gf.value AS EN_ATM_CO2E_GF_KT
FROM all_dates d
LEFT JOIN country_data co2_tot
    ON co2_tot.country_code = 'USA'
    AND co2_tot.indicator_code = 'EN_ATM_CO2E_KT'
    AND co2_tot.date = d.date
LEFT JOIN country_data co2_pc
    ON co2_pc.country_code = 'USA'
    AND co2_pc.indicator_code = 'EN_ATM_CO2E_PC'
    AND co2_pc.date = d.date
LEFT JOIN country_data co2_sf
    ON co2_sf.country_code = 'USA'
    AND co2_sf.indicator_code = 'EN_ATM_CO2E_SF_KT'
    AND co2_sf.date = d.date
LEFT JOIN country_data co2_lf
    ON co2_lf.country_code = 'USA'
    AND co2_lf.indicator_code = 'EN_ATM_CO2E_LF_KT'
    AND co2_lf.date = d.date
LEFT JOIN country_data co2_gf
    ON co2_gf.country_code = 'USA'
    AND co2_gf.indicator_code = 'EN_ATM_CO2E_GF_KT'
    AND co2_gf.date = d.date
ORDER BY d.date;

select * from M6;

-- M7: Calculate the average percentage growth for a chosen indicator per country
↪ from 2010 onwards.
-- This view computes the average growth rate and orders countries by the highest
↪ growth.

CREATE OR REPLACE VIEW M7 AS
WITH yearly AS (
    SELECT
        cd.country_code,
        c.name AS country_name,
        cd.date,

```

```

        cd.value,
        LAG(cd.value) OVER (
            PARTITION BY cd.country_code
            ORDER BY cd.date
        ) AS prev_value
FROM country_data cd
JOIN country c ON c.code = cd.country_code
WHERE cd.indicator_code = 'EN_ATM_CO2E_KT'
      AND cd.date >= DATE '2010-01-01'
),
growth AS (
    SELECT
        country_name,
        100.0 * (value - prev_value) / prev_value AS pct_growth
    FROM yearly
    WHERE prev_value IS NOT NULL
          AND prev_value <> 0
)
SELECT
    country_name,
    ROUND(AVG(pct_growth)::numeric, 2) AS avg_pct_growth
FROM growth
GROUP BY country_name
ORDER BY avg_pct_growth DESC;

Select * from M7;

-- M8: Identify countries that have data for any selected indicator in 2011.
-- This view finds countries with available data in the year 2011.

CREATE OR REPLACE VIEW M8 AS
SELECT c.name AS country_name
FROM country c
WHERE EXISTS (
    SELECT *
    FROM country_data cd
    WHERE cd.country_code = c.code
          AND cd.indicator_code = 'EN_ATM_CO2E_SF_KT'

```

```

        AND EXTRACT(YEAR FROM cd.date) = 2011
    )
ORDER BY c.name;

Select * from M8;

-- M9: Find countries that have data for all selected indicators in 2011.
-- This view ensures that countries listed have complete data for the year 2011.

CREATE OR REPLACE VIEW M9 AS
SELECT
    c.name AS country_name
FROM country c
JOIN country_data cd
    ON cd.country_code = c.code
WHERE cd.indicator_code IN (
    'EN_ATM_CO2E_KT',
    'EN_ATM_CO2E_PC',
    'EN_ATM_CO2E_LF_KT',
    'EN_ATM_CO2E_SF_KT',
    'EN_ATM_CO2E_GF_KT'
)
    AND cd.date >= DATE '2011-01-01'
    AND cd.date < DATE '2012-01-01'
GROUP BY c.name
HAVING COUNT(DISTINCT cd.indicator_code) = 5
ORDER BY c.name;

SELECT * FROM M9;

-- A1: Examine the relationship between GDP per capita and internet adoption.
-- This view pairs GDP per capita with internet users per 100 people for each
    ↪ country and calculates their averages.
CREATE OR REPLACE VIEW A1 AS
WITH paired AS (
    SELECT
        d_gdp.country_code,
        c.name AS country_name,

```

```

        d_gdp.date,
        d_gdp.value AS gdp_pc,
        d_net.value AS internet_users
FROM country_data d_gdp
JOIN country_data d_net
    ON d_gdp.country_code = d_net.country_code
    AND d_gdp.date = d_net.date
JOIN country c ON c.code = d_gdp.country_code
WHERE d_gdp.indicator_code = 'NY_GDP_PCAP_CD'
    AND d_net.indicator_code = 'IT_NET_USER_P2'
)
SELECT
    country_name,
    ROUND(AVG(gdp_pc)::numeric, 2) AS avg_gdp_pc,
    ROUND(AVG(internet_users)::numeric, 2) AS avg_internet_users
FROM paired
GROUP BY country_name
HAVING COUNT(*) >= 5
ORDER BY avg_gdp_pc DESC;

Select * from A1;

-- A2: Analyze the relationship between internet adoption and CO2 emissions per
↪ capita.
-- This view pairs internet usage with CO2 emissions per capita to reveal
↪ potential correlations.
CREATE OR REPLACE VIEW A2 AS
WITH paired AS (
    SELECT
        d_net.country_code,
        c.name AS country_name,
        d_net.date,
        d_net.value AS internet_users,
        d_co2.value AS co2_pc
    FROM country_data d_net
    JOIN country_data d_co2
        ON d_net.country_code = d_co2.country_code
        AND d_net.date = d_co2.date

```

```

        JOIN country c ON c.code = d_net.country_code
        WHERE d_net.indicator_code = 'IT_NET_USER_P2'
              AND d_co2.indicator_code = 'EN_ATM_CO2E_PC'
    )
SELECT
    country_name,
    ROUND(AVG(internet_users)::numeric, 2) AS avg_internet_users,
    ROUND(AVG(co2_pc)::numeric, 2) AS avg_co2_pc
FROM paired
GROUP BY country_name
HAVING COUNT(*) >= 5
ORDER BY avg_internet_users DESC;

Select * From A2;

-- A3: Compare GDP growth and CO2 growth to analyze decoupling.
-- This view calculates and compares the average percentage growth of GDP and CO2
↪ emissions for each country.

CREATE OR REPLACE VIEW A3 AS
WITH yearly AS (
    SELECT
        country_code,
        date,
        indicator_code,
        value,
        LAG(value) OVER (
            PARTITION BY country_code, indicator_code
            ORDER BY date
        ) AS prev_val
    FROM country_data
    WHERE indicator_code IN ('NY_GDP_PCAP_CD', 'EN_ATM_CO2E_PC')
        AND date >= DATE '2010-01-01'
),
growth AS (
    SELECT
        country_code,
        indicator_code,

```

```

        100.0 * (value - prev_val) / prev_val AS pct_growth
    FROM yearly
    WHERE prev_val IS NOT NULL AND prev_val <> 0
),
avg_growth AS (
    SELECT
        country_code,
        indicator_code,
        AVG(pct_growth) AS avg_pct_growth
    FROM growth
    GROUP BY country_code, indicator_code
)
SELECT
    c.name AS country_name,
    ROUND(gdp.avg_pct_growth::numeric, 2) AS gdp_growth,
    ROUND(co2.avg_pct_growth::numeric, 2) AS co2_growth,
    ROUND((co2.avg_pct_growth - gdp.avg_pct_growth)::numeric, 2) AS
    ↪ decoupling_gap
FROM avg_growth gdp
JOIN avg_growth co2 ON gdp.country_code = co2.country_code
JOIN country c ON c.code = gdp.country_code
WHERE gdp.indicator_code = 'NY_GDP_PCAP_CD'
    AND co2.indicator_code = 'EN_ATM_CO2E_PC'
ORDER BY decoupling_gap ASC;

Select * from A3;

-- A4: Examine the relationship between inflation and GDP per capita.
-- This view pairs inflation rates with GDP per capita to explore economic
    ↪ stability and development.

CREATE OR REPLACE VIEW A4 AS
WITH paired AS (
    SELECT
        d_inf.country_code,
        c.name AS country_name,
        d_inf.date,
        d_inf.value AS inflation,

```

```

        d_gdp.value AS gdp_pc
FROM country_data d_inf
JOIN country_data d_gdp
    ON d_inf.country_code = d_gdp.country_code
    AND d_inf.date = d_gdp.date
JOIN country c ON c.code = d_inf.country_code
WHERE d_inf.indicator_code = 'FP_CPI_TOTL_ZG'
    AND d_gdp.indicator_code = 'NY_GDP_PCAP_CD'
)
SELECT
    country_name,
    ROUND(AVG(inflation)::numeric, 2) AS avg_inflation,
    ROUND(AVG(gdp_pc)::numeric, 2) AS avg_gdp_pc
FROM paired
GROUP BY country_name
HAVING COUNT(*) >= 5
ORDER BY avg_gdp_pc DESC;

Select * from A4;

-- A5: Analyze fuel-type emissions by development level.
-- This view shows the most recent values of different fuel-type CO2 emissions
↪ alongside GDP per capita for each country.

CREATE OR REPLACE VIEW A5 AS
WITH latest_gdp AS (
    SELECT
        country_code,
        value AS gdp_pc,
        ROW_NUMBER() OVER (
            PARTITION BY country_code
            ORDER BY date DESC
        ) AS rn
    FROM country_data
    WHERE indicator_code = 'NY_GDP_PCAP_CD'
),
latest_energy AS (
    SELECT

```

```

        country_code,
        indicator_code,
        value,
        ROW_NUMBER() OVER (
            PARTITION BY country_code, indicator_code
            ORDER BY date DESC
        ) AS rn
FROM country_data
WHERE indicator_code IN (
    'EN_ATM_CO2E_SF_KT',
    'EN_ATM_CO2E_LF_KT',
    'EN_ATM_CO2E_GF_KT'
)
)
)
SELECT
    c.name AS country_name,
    ROUND(g.gdp_pc::numeric, 2) AS gdp_per_capita,
    ROUND(sf.value::numeric, 2) AS solid_fuel_co2_kt,
    ROUND(lf.value::numeric, 2) AS liquid_fuel_co2_kt,
    ROUND(gf.value::numeric, 2) AS gas_fuel_co2_kt,
    ROUND( (COALESCE(sf.value,0) + COALESCE(lf.value,0) +
↪ COALESCE(gf.value,0))::numeric , 2)
        AS total_fuel_co2_kt
FROM latest_gdp g
JOIN country c ON c.code = g.country_code
LEFT JOIN latest_energy sf
    ON sf.country_code = g.country_code
    AND sf.indicator_code = 'EN_ATM_CO2E_SF_KT'
    AND sf.rn = 1
LEFT JOIN latest_energy lf
    ON lf.country_code = g.country_code
    AND lf.indicator_code = 'EN_ATM_CO2E_LF_KT'
    AND lf.rn = 1
LEFT JOIN latest_energy gf
    ON gf.country_code = g.country_code
    AND gf.indicator_code = 'EN_ATM_CO2E_GF_KT'
    AND gf.rn = 1
WHERE g.rn = 1

```



```

    AND (sf.value IS NOT NULL OR lf.value IS NOT NULL OR gf.value IS NOT NULL)
ORDER BY g.gdp_pc DESC;

Select * From A5;

/* Mandatory Analysis
Report for PDF M1 - M9: Select statements to call each view Limit to 10 Rows */

-- M1: Countries missing data for selected indicators (limit to 10 rows)
SELECT * FROM M1
LIMIT 10;

-- M2: Countries with complete data for all indicators (limit to 10 rows)
SELECT * FROM M2
LIMIT 10;

-- M3: Indicators with the number of countries with data, sorted by count (limit
↪ to 10 rows)
SELECT * FROM M3
LIMIT 10;

-- M4: Top 10 countries by most recent CO2 emissions (limit to 10 rows)
SELECT * FROM M4
LIMIT 10;

-- M5: Average CO2 emissions per capita by continent (limit to 10 rows)
SELECT * FROM M5
LIMIT 10;

-- M6: U.S. indicator data over time (limit to 10 rows)
SELECT * FROM M6
LIMIT 10;

-- M7: Average growth percentage for each country (limit to 10 rows)
SELECT * FROM M7
LIMIT 10;

-- M8: Countries with 2011 data for any indicator (limit to 10 rows)

```

```

SELECT * FROM M8
LIMIT 10;

-- M9: Countries with complete 2011 data for all indicators (limit to 10 rows)
SELECT * FROM M9
LIMIT 10;

/* Additional Analysis
Report for PDF A1 - A5: Select statements to call each view Limit to 10 Rows */

-- A1: Development vs Technology Adoption (limit to 10 rows)
SELECT * FROM A1
LIMIT 10;

-- A2: Technology vs Environmental Impact (limit to 10 rows)
SELECT * FROM A2
LIMIT 10;

-- A3: Economic vs Environmental Growth (limit to 10 rows)
SELECT * FROM A3
LIMIT 10;

-- A4: Economic Stability vs Development (limit to 10 rows)
SELECT * FROM A4
Limit 10;

-- A5: Energy Mix by Development Level (limit to 10 rows)
SELECT * FROM A5
LIMIT 10;

```